The Nouns Beneath the Numbers

Ontological Fluency and the Difficulty of School Physics

A reflective essay on misconceptions, symbolic operations and the hidden architecture of physical quantities

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Students may fail to solve physics problems even when they remember the formula and can perform the algebra. The difficulty can lie deeper: they may not know what kind of thing a symbol represents, what relation it expresses, or what operation the equation authorises. Physics problem solving therefore requires more than symbolic fluency. It requires ontological fluency.

School physics is densely populated by nouns. Force, mass, energy, current, field, resistance, acceleration, density, momentum and power enter the classroom as though each named a stable item in the inventory of nature.

The grammar is reassuring. A noun appears to point towards a thing. A definition appears to tell us what that thing is. An equation appears to state how the named things are related. The student is then expected to substitute values, rearrange symbols and arrive at an answer.

Yet the apparent uniformity is misleading. These terms do not all denote the same kind of entity.

A force is an interaction represented by a vector. Acceleration is a rate of change of velocity. Density is a ratio property. Current is a rate of charge transfer. Power is a rate of energy transfer. A field assigns a quantity to locations in space and time. Resistance is a relational parameter within a circuit model. Energy functions as a conserved quantity within a defined system.

To place all of these under the grammatical heading of "things" is already to hide much of the physics.

This essay develops a simple proposition:

> Many persistent errors in school physics are not merely failures of recall or algebra. They are failures to identify what kind of entity, process, relation or rate a symbol represents.

Such failures may be described as **ontological misclassifications**. The corresponding educational capacity will be called **ontological fluency**: the ability to recognise what kind of quantity or relation a physics term denotes, how that status depends on the model being used, and which mathematical operations are licensed by it.

A companion paper, *Ontological Fluency in Physics: Diagnostic and Classroom Tasks*, develops the diagnostic questions and classroom activities proposed here. The present essay establishes the conceptual argument.

The ontology–operation gap

Consider the familiar equation

$$F = ma.$$

A student may read it as a compressed sentence:

> Force is mass times acceleration.

This sentence seems simple because it converts three symbols into three familiar nouns. But little has yet been understood.

What is force here? Is it an object, a property possessed by an object, a cause, an interaction, or the numerical result of multiplying two other quantities?

What is acceleration? Is it a force-like thing acting on the object, a property carried by the object, or a description of how velocity changes relative to a frame?

What does the equality sign mean? Does mass combine with acceleration to produce force, as ingredients produce a mixture? Does force cause acceleration? Or does the equation constrain three quantities within a specified model?

The more physically explicit forms are

$$F_{\text{net}} = ma,$$

$$\sum \mathbf{F} = m\mathbf{a},$$

and, more generally,

$$\sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}.$$

Each version clarifies something. The relevant force is a net force. Several interactions may contribute. The quantities are vectors. The constant-mass form is a special case of the momentum relation.

Yet every clarification increases the visible symbolic burden. The subscript, summation sign, vector arrow, index and derivative all add elements that must be interpreted.

The simpler expression therefore does not simply remove difficulty. It moves part of the difficulty off the page. The expert supplies the missing system boundary, vector character, interaction structure and modelling conditions from an organised conceptual schema. The novice encounters three letters whose ontological roles remain uncertain.

> **"The simpler equation may remove notation by transferring its work into interpretation."**

This is the **ontology–operation gap**. Students are asked to operate on symbols before they have securely classified what the symbols represent.

From things to processes

Chi, Slotta and de Leeuw proposed that some scientific misconceptions persist because a learner assigns a scientific concept to an inappropriate ontological category [1]. A student may treat a process as though it were matter, or a constraint-based interaction as though it were a substance moving from place to place.

Conceptual change is then more demanding than correcting an isolated belief. The learner must relocate the concept into another category.

Reiner, Slotta, Chi and Resnick developed this argument through examples of naive physics reasoning in which heat, light, force and electric current are treated in substance-like ways [2]. Such reasoning is not necessarily incoherent. It can support a stable, internally consistent interpretation of a phenomenon.

If current is imagined as a material substance, then it is reasonable to expect some of it to be consumed by a bulb. If force is imagined as an impetus placed inside a moving body, then it is reasonable to expect the body to stop once the impetus has been used up. If heat is imagined as a fluid held by an object, then it is reasonable to speak of heat being stored, lost or contained.

The scientific error is real. The reasoning that produces it may nonetheless be coherent within the student's ontology.

That observation changes the diagnostic question. Instead of asking only

> Which fact has the student forgotten?

we should also ask

> What must the student believe this quantity *is* for the answer to make sense?

Nominalisation and reification

Scientific language depends heavily on nominalisation: the transformation of a verb, process or relation into a noun.

A body accelerates. We speak of its acceleration.

Energy transfers. We speak of an energy transfer.

A charge distribution interacts with other charges. We speak of an electric field and an electric force.

Nominalisation is indispensable. It permits processes to become objects of thought. Once named, they can be compared, quantified, placed into equations and related to other concepts.

The danger begins when the noun is reified: when an abstraction or relation is treated as a concrete object, substance or agent.

The progression can be subtle:

$$\begin{aligned}\text{the body accelerates} &\longrightarrow \text{the body has acceleration,}\\ &\longrightarrow \text{acceleration acts on the body.}\end{aligned}$$

Or:

$$\begin{aligned}\text{the velocity remains constant} &\longrightarrow \text{the body has inertia,}\\ &\longrightarrow \text{inertia keeps it moving.}\end{aligned}$$

The noun acquires agency.

Wittgenstein's later philosophy is useful here because it directs attention towards how a word functions within a practice rather than towards an imagined essence concealed behind the word [8]. The surface grammar of a statement can make a redescription look like an explanation.

"The object continues because of inertia" may appear explanatory. Yet if "inertia" means only the tendency expressed by Newton's first law, the statement risks explaining continued motion by renaming continued motion.

The classroom phrase has not necessarily revealed a mechanism. It may have transformed a pattern into a noun and then assigned the noun causal power.

Five forms of ontological misclassification

The misconceptions considered here can be organised into five broad families. These categories overlap, but the distinctions are useful for diagnosis.

1. The substance error

A process, rate or bookkeeping quantity is treated as material stuff.

Common examples include:

> Heat is stored in an object.
>
> Electricity flows through the wire.
>
> Current is used up by the bulb.
>
> Energy is contained in the battery.

The language of storage, flow, possession and consumption is often productive. It supports diagrams, conservation reasoning and simple calculations. It also encourages students to imagine an underlying substance whose movement is the phenomenon.

Consider current:

$$I = \frac{\Delta Q}{\Delta t}.$$

Current is not a quantity of charge carried inside a component. It is a rate associated with charge crossing a boundary. In a steady series circuit, the same rate occurs at each point, although energy is transferred in the components.

A student who predicts

$$I_{\text{before bulb}} > I_{\text{after bulb}}$$

may be preserving a coherent substance ontology: some current entered the bulb, some was consumed, and less remained afterwards.

The correction "current is the same everywhere" may alter the answer without altering the ontology. The underlying image can remain available and reappear in another context.

2. The agent error

A theoretical term is treated as an actor that makes events happen.

Examples include:

> Gravity pulls the object down.
>
> The battery pushes electrons around the circuit.
>
> Heat seeks equilibrium.

Inertia keeps the object moving.

Light chooses the path of least time.

These phrases compress difficult relations into accessible verbs. Their pedagogical usefulness is therefore genuine. The problem arises when metaphorical agency is mistaken for mechanism.

"Gravity pulls" may function adequately in a near-Earth Newtonian model. It becomes less secure when the student must distinguish gravitational force from gravitational field strength, compare inertial and gravitational mass, or encounter the geometric description of gravitation in general relativity.

Similarly, "the battery pushes electrons" can help initiate a circuit discussion. It can also conceal the roles of potential difference, electric field, charge-carrier mobility and energy transfer.

The agent error is especially likely to generate spurious forces. Students may draw a forward "force of motion" on a coasting object or an "inertia force" that keeps it moving. Motion is then treated as evidence of a sustaining agent.

3. The attribute error

A relational quantity is treated as an intrinsic possession of a single object.

School questions frequently use the grammar of possession:

The object has a mass of 5 kg.

The object has a force of 20 N.

The wire has a current of 3 A.

The charge has an electric field.

These sentences have similar grammatical structure while expressing different physical relations.

Mass can often be treated as a property of the selected system. A force, however, is force *on* one system *by* another. Current is not possessed by an isolated wire in the same sense that the wire possesses a length. An electric field is assigned to locations and associated with a source configuration; it is not simply an object carried by a charge.

The flattened predicate "has" removes these distinctions.

A subject–predicate–object representation can restore them. Instead of

$$\text{object} \longrightarrow \text{has} \longrightarrow \text{force},$$

write

$$\text{Earth} \longrightarrow \text{exerts gravitational force on} \longrightarrow \text{object}.$$

Instead of

$$\text{wire} \longrightarrow \text{has} \longrightarrow \text{current},$$

write

$$\text{charge} \longrightarrow \text{crosses section per unit time} \longrightarrow \text{current}.$$

The first graph makes an interaction visible. The second transforms a possessed object into a rate relation.

Knowledge graphs are therefore more than organisational devices. Their edges disclose the ontological assumptions hidden by classroom grammar. A graph can show whether a quantity is being represented as a property, an interaction, a measurement, an inference or a theoretical construct.

4. The process–state error

A state-like quantity is confused with a rate, change or transfer.

Several familiar pairs have this structure:

velocity and acceleration,

energy and power,

charge and current,

temperature and heat transfer.

The corresponding equations include

$$a = \frac{\Delta v}{\Delta t}, \qquad P = \frac{\Delta E}{\Delta t}, \qquad I = \frac{\Delta Q}{\Delta t}.$$

The common mathematical form is a quotient. The physical meanings differ, but each requires the learner to interpret a numerator as changing, transferring or crossing during an interval.

A student may manipulate the formula correctly while retaining an incorrect ontology. They may calculate power while imagining it as an amount of energy; calculate current while imagining it as accumulated charge; or calculate acceleration while imagining it as increasing speed only.

School algebra often presents the fraction as an operation to perform. Physics requires the quotient to be read as a rate with a temporal, spatial or systemic interpretation.

The student must therefore learn two readings simultaneously:

$$\frac{\Delta E}{\Delta t} \quad \text{as division,}$$

and

$$\frac{\Delta E}{\Delta t} \quad \text{as energy transferred per unit time.}$$

The first is an algebraic operation. The second is a physical ontology.

5. The model–object error

A representation is mistaken for the phenomenon represented.

Examples include:

Magnetic field lines emerge from one pole and enter the other.

A ray traces the route travelled by a photon.

Particles in an ideal gas are tiny hard spheres exactly as drawn.

A force arrow is a visible feature attached to the object.

An ideal gas obeys the gas law, while a real gas disobeys it.

Field lines are visual constructions. Rays are geometrical representations. Ideal gases are model systems. Force arrows encode magnitude and direction relative to a chosen representation.

The drawings are not dispensable decorations. They make reasoning possible. Yet their usefulness depends upon a disciplined distinction between the model and the world.

The phrase "real gases deviate from ideal behaviour" is revealing. The real gas is grammatically treated as a poorly behaved imitation of the ideal model. In fact, it is the model that ceases to approximate the gas adequately under certain conditions.

The direction of comparison has been reversed.

Density and the hidden complexity of a ratio

Density provides a modest but revealing example:

$$\rho = \frac{m}{V}.$$

The arithmetic is elementary. Density often appears in mathematics lessons through measurement, unit conversion and compound measures. Students may already know that one divides mass by volume.

They may also have heard density described repeatedly as

> the amount of stuff packed into a given volume.

The description is memorable. It does not by itself establish fluency.

The symbol ρ is unfamiliar. The word "stuff" is ontologically vague. The word "packed" suggests particles being compressed into a container. The phrase "amount in a volume" can encourage the learner to see density as the amount itself rather than as a ratio property.

To use the equation reliably, the student must coordinate several commitments:

1. m is the mass of a selected sample;
2. V is the volume occupied by that same sample;
3. ρ is the ratio of those measurements;
4. for a homogeneous material, the ratio may be treated as a material property over the relevant range;
5. the units must preserve the dimensional relation $[\rho] = ML^{-3}$.

A student who inserts the volume of one body and the mass of another has not merely made an arithmetic mistake. The student has failed to maintain the identity of the system across the quantities.

A student who says that a larger object must have greater density may be confusing an extensive quantity, mass, with an intensive ratio property.

A student who treats ρ as an additional substance inside the object may understand the equation as a recipe in which mass and volume produce a third ingredient.

The equation is short. Its ontological commitments are not.

Reversal errors and noun-label algebra

Clement's classic reversal-error study showed that even relatively advanced students could experience serious difficulty translating verbal relationships into algebraic equations [3].

Given a statement such as

> There are six times as many students as professors,

students may write

$$6S = P$$

instead of

$$S = 6P.$$

One source of the error is syntactic word-order matching. Another is a static-comparison interpretation in which the coefficient is attached to the larger group.

The letters can function as compressed noun labels:

$$S \equiv \text{students}, \qquad P \equiv \text{professors},$$

rather than as variables representing numbers of students and professors.

Physics equations invite the same mistake. The letter F may be read as the word "force"; m as the word "mass"; and a as the word "acceleration". The equation then resembles a sentence assembled from labelled nouns.

But a physics symbol does not merely abbreviate a word. It represents a quantity belonging to a system, carrying dimensions and units, embedded in a model, and often possessing direction or dependence on other variables.

To read

$$F = ma$$

as "force equals mass times acceleration" is therefore only the beginning of interpretation. It does not establish which force, whose mass, which component of acceleration, relative to which frame, or under what conditions the relation applies.

Equations as symbolic forms

Sherin's account of **symbolic forms** provides a bridge between mathematical structure and conceptual meaning [4]. A symbolic form connects a recurring pattern in an equation with an intuitive conceptual schema.

Experts do not normally decode every symbol independently. They recognise structures.

In

$$\rho = \frac{m}{V},$$

the expert sees a whole distributed over a region, or an amount per unit extent.

In

$$\sum_i \mathbf{F}_i = m\mathbf{a},$$

the expert sees several contributions accumulating into a resultant that constrains change of motion.

In

$$V = IR,$$

the expert can see a proportionality involving a system parameter, not merely three letters in a multiplication pattern.

The equation has become a conceptual object.

This helps explain why a notation that appears complicated to a novice can appear economical to an expert. The expert has chunked the form and its meaning together. The novice must coordinate separate symbols, definitions, units and operations.

Redish argues that mathematics in physics functions as a disciplinary dialect in which formal symbolism and physical interpretation are blended [5]. Kuo and colleagues similarly describe successful problem solving as an opportunistic blending of conceptual and formal reasoning, including during algebraic manipulation rather than only before or after it [6].

Torigoe's work on unpacking symbolic equations further emphasises that symbolic and numerical equations afford different kinds of reasoning [7]. A symbolic equation represents a class of systems. Its apparent indeterminacy is part of its power.

The educational problem is therefore larger than formula recall. Students must learn how an equation can be read, unpacked, specialised, tested and reconstructed.

From phenomenon to operation

A physics problem can be represented as a sequence:

$$\text{phenomenon} \longrightarrow \text{system} \longrightarrow \text{entities} \longrightarrow \text{relations} \longrightarrow \text{quantities} \longrightarrow \text{equation} \longrightarrow \text{operation}.$$

Much routine instruction begins near the end:

$$\text{quantities} \longrightarrow \text{equation} \longrightarrow \text{operation}.$$

Values are extracted from the question. A formula is selected. The formula is rearranged. The numbers are substituted.

The method can be successful when the problem closely resembles a learned example. Transfer becomes fragile because the earlier modelling decisions remain implicit.

Before applying Newton's second law, the solver must decide:

- what system is being analysed;
- which interactions cross its boundary;
- which quantities are vectors;
- which direction or coordinate convention will be used;
- whether the frame may be treated as inertial;
- whether the mass is constant;
- whether the acceleration concerns speed, direction, or both.

These are ontological and representational decisions. They determine what the algebra means.

A formula sheet can remove the requirement to recall the final equation. It cannot perform the prior acts of system selection, classification and model construction.

Misconceptions as alternative ontologies

The word "misconception" can suggest a defective proposition waiting to be replaced by a correct one. Some errors are better understood as consequences of a coherent alternative ontology.

Consider four common responses.

A moving object must have a forward force

If force is understood as a motive property possessed by the body, motion is evidence that the force remains inside it. The answer follows naturally.

The scientific account instead distinguishes velocity from acceleration and force from motion:

$$\sum \mathbf{F} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{a} = \mathbf{0},$$

while

$\mathbf{v}$

may remain non-zero and constant.

The conceptual repair requires more than stating Newton's first law. Force must be reclassified from a stored property of motion to an interaction contributing to change of momentum.

A heavier body falls faster because gravity is stronger

The student may correctly recognise that

$$F_g = mg$$

is larger for a larger mass. The conclusion follows if force alone is treated as the determinant of acceleration.

Newton's second law supplies the inertial response:

$$a = \frac{F_g}{m} = \frac{mg}{m} = g.$$

The cancellation is algebraic, but its meaning is ontological. The same mass that increases gravitational force also measures inertial response within the Newtonian model.

Current is used up in a bulb

If current is a substance, transfer of energy appears to require consumption of the carrier. The scientifically relevant distinction is

$$I = \frac{\Delta Q}{\Delta t}$$

as charge-transfer rate, while

$$P = VI$$

describes energy-transfer rate.

The bulb transfers energy without consuming steady current.

Power is the energy an appliance has

The formula

$$P = \frac{E}{t}$$

may be manipulated correctly while power is still treated as an amount. The student must reclassify power as a rate, with watts understood as joules per second.

In each case, the wrong answer can be rational within the student's category system. Correction requires a shift in what the quantity is taken to be.

Ontological fluency

We may now define **ontological fluency** more precisely:

> Ontological fluency is the capacity to identify whether a physics term is functioning as an object, property, relation, interaction, state, process, rate, field, model parameter or representational construct, and to use that classification appropriately when interpreting and manipulating equations.

This does not require students to adopt a final metaphysical position on whether fields or forces are "really real". The classroom aim is more operational.

Students should recognise, for example, that:

- force is relational and vectorial;
- acceleration is a rate of change relative to a frame;
- current and power are rates;
- density is a ratio property;
- energy is defined relative to a selected system and model;
- a field assigns quantities to locations;
- a field line is a representation rather than a physical filament;
- an ideal gas is a model system rather than a perfect real gas.

Ontological fluency should be distinguished from two neighbouring forms of competence.

Symbolic fluency concerns the ability to read, rearrange and manipulate mathematical expressions.

Model-selection fluency concerns the ability to decide which idealisation, equation or representation applies to a situation.

The relationship may be expressed heuristically as

$$\begin{aligned}\text{physics problem-solving fluency} \approx\ & \text{symbolic fluency}\\ & + \text{ontological fluency}\\ & + \text{model-selection fluency}.\end{aligned}$$

This is not a psychometric equation. It is a reminder that algebraic competence alone does not constitute physical problem-solving competence.

The classroom programme

The companion paper develops a classroom programme from this framework. Its tasks are grouped into five families.

Classifying the noun

Students classify terms such as

$$\text{mass, force, acceleration, density, current, energy, field}$$

using categories such as

$$\text{property, relation, rate, interaction, process, model parameter.}$$

The discussion matters more than the production of a fixed taxonomy. Students must justify which mathematical and representational operations their classification permits.

Repairing the sentence

Students rewrite statements whose grammar conceals physical relations.

For example:

> The object has a force of 20 N.

becomes

> The rope exerts a 20 N force on the object.

And:

> The battery pushes current through the resistor.

becomes

> The battery maintains a potential difference; an electric field in the circuit produces charge-carrier drift.

The purpose is not to prohibit metaphor. It is to teach students to move between compact classroom language and more explicit model language.

Expanding the equation into a graph

Students unpack an equation into subject–predicate–object relations.

For

$$F_{\text{net}} = ma,$$

a graph might include

$$\text{Earth} \longrightarrow \text{exerts gravitational force on} \longrightarrow \text{object},$$

$$\text{surface} \longrightarrow \text{exerts normal force on} \longrightarrow \text{object},$$

and

$$\text{vector sum of external interactions} \longrightarrow \text{constrains} \longrightarrow \text{acceleration}.$$

This activity links knowledge graphs with free-body diagrams and equation construction.

Diagnosing the incorrect solution

Students are shown an incorrect response and asked:

> What must this solver believe the quantity is for this answer to make sense?

The question shifts attention from error correction to conceptual reconstruction.

Moving between representations

Students translate among:

$$\text{phenomenon, sentence, diagram, graph, equation, units.}$$

Each representation reveals some commitments and suppresses others. Fluency consists partly in knowing what has been made explicit and what must still be supplied.

The full tasks, teacher notes and suggested diagnostic responses are published in the companion document:

Ontological Fluency in Physics: Diagnostic and Classroom Tasks.

The Ontology Atlas, an interactive map of the same relations.

The companion workbook develops these diagnostic questions, sentence repairs, knowledge graphs and equation-unpacking activities as classroom tasks; the accompanying atlas holds the same structure as a navigable graph.

Ontology and cognitive load

This framework also sharpens the earlier criticism of cognitive load as educational phlogiston.

Cognitive load is a genuine constraint. A learner cannot indefinitely maintain and coordinate separate symbols, definitions, diagrams, units and conditions.

Yet saying that a student finds

$$\rho = \frac{m}{V}$$

difficult because it imposes excessive cognitive load leaves open the central question: why do the elements remain separate?

For the expert, the expression may function as one conceptual unit:

> an amount distributed through a volume.

For the novice, it may remain three symbols, a division sign, an unfamiliar Greek letter, two measurements and several possible units.

Similarly, the expert may read

$$\sum_i \mathbf{F}_i = m\mathbf{a}$$

as a single system-level relationship between external interactions and change of motion. The novice may encounter a summation, index, vectors, mass, acceleration and an unexplained selection of forces.

The experienced load differs because the conceptual organisation differs.

Ontological fluency helps explain what has been chunked. It identifies the category structure that allows several symbols and operations to become one meaningful form.

Thus:

> Cognitive load describes the burden of coordinating the elements. Ontological fluency helps explain whether those elements can cohere.

Without the second analysis, cognitive load can become a label attached after meaning has failed.

Teaching truths, tools and disciplined fictions

The argument does not require that every scientific term be treated as a mere fiction.

A scientific realist may regard fields, particles and forces as referring to genuine structures or entities in the world. A model-based philosopher may emphasise the domain-bound function of these concepts. A Wittgensteinian analysis may focus on the practices that establish their meaning.

The educational problem occurs before these positions need to be settled.

Students must know how the concept functions in the model being used.

A magnetic field may be discussed as physically real, as a mathematical assignment to space, or as part of an effective model. In every case, field lines must not be confused with material threads.

Energy may be described realistically or instrumentally. In every case, students must distinguish energy from power and internal energy from heat transfer.

Force may be treated as an interaction, a model quantity or a causal relation. In every case, it must not be confused with velocity or stored motion.

The aim is therefore a form of disciplined pluralism. Students should be able to use a representation without mistaking it for the phenomenon, use a metaphor without mistaking it for a mechanism, and use a noun without assuming that the noun names a substance.

Conclusion: the nouns beneath the numbers

Physics equations are often presented as the point at which language has become precise.

Yet the symbols inherit the ambiguities of the words beneath them.

The letter F can conceal an interaction.

The letter a can conceal a changing vector.

The letter I can conceal a rate across a boundary.

The symbol ρ can conceal a ratio property.

The letter E can conceal a bookkeeping quantity defined relative to a system.

The algebra does not automatically resolve these ontological questions. It compresses them.

This helps explain why students can remember a formula, substitute values correctly in familiar exercises and still fail when the surface context changes. They have learned an operation without acquiring the conceptual classification that gives the operation physical meaning.

The resulting errors are not always random. They may be coherent consequences of treating rates as substances, interactions as possessions, models as objects, or abstract nouns as agents.

Physics problem solving therefore begins before the equation is selected. It begins with the construction of a world in which the equation can mean something.

The learner must decide:

what the system is,
what entities are present,
what relations connect them,
what quantities represent those relations,
and what operations those quantities permit.

Only then does symbolic manipulation become physical reasoning.

Ontological fluency names this neglected capacity. It offers a route from philosophical reflection to classroom diagnosis, from the grammar of scientific nouns to the practical errors found in student calculations.

Cognitive load may tell us that too much is being coordinated.

Ontological analysis asks the more demanding question:

> What kind of things does the student believe they are coordinating?

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A companion task paper will develop the diagnostic questions, sentence repairs, knowledge graphs and equation-unpacking activities outlined here.